



# UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 01087

LM NUCLEAR PHYSICS

SUMMER EXAMINATIONS 2025

*Time allowed: 2 hours*

***Answer four questions from Section 1 and two questions from Section 2.***

Section 1 consists of four questions and carries 40% of the marks for the examination.

Answer ***all four*** questions from this section.

Section 2 consists of three questions and carries 60% of the marks.

Answer ***two*** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [ ].

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

## SECTION 1

Answer **all four** questions from this section.

1. (a) Write down the expression for the Woods-Saxon potential, explaining the meaning of the parameters, and sketch the potential as a function of radius.

[6]

- (b) Explain where the shell model does a poor job at predicting the spin-parities of ground states. What phenomena make some of the underlying assumptions in the shell model inaccurate?

[4]

2. For each of the  $\beta$ -decay transitions below, note whether the transition is Fermi, Gamow-Teller, or mixed, as well as whether it is allowed or forbidden (and what order).

(a)  $^{13}\text{N} (J^\pi = \frac{1}{2}^-) \rightarrow ^{13}\text{C} (J^\pi = \frac{1}{2}^-)$

(b)  $^{14}\text{O} (J^\pi = 0^+) \rightarrow ^{14}\text{N}^* (J^\pi = 0^+)$

(c)  $^{40}\text{K} (J^\pi = 4^-) \rightarrow ^{40}\text{Ca} (J^\pi = 0^+)$

[10]

3. The cross section for the reaction:

$$\bar{\nu}_e + p \rightarrow n + e^+,$$

is  $1.2 \times 10^{-19}$  barns.

- (a) Calculate the reaction rate from this process through a 10 cm thick target of water ( $\text{H}_2\text{O}$ ) which has a density of  $1 \text{ g/cm}^3$  from a neutrino beam of  $10^9$  particles per second.

[8]

- (b) Which of the four fundamental forces mediates this interaction and what are the possible exchange bosons involved?

[2]

**ANY CALCULATOR**

4. A  $J^\pi = 0^+$  state can decay by emitting 3 possible gamma rays, with the following measured energies and multipolarities:

- 2.430 MeV, E3
- 1.115 MeV, E2
- 1.315 MeV, E1

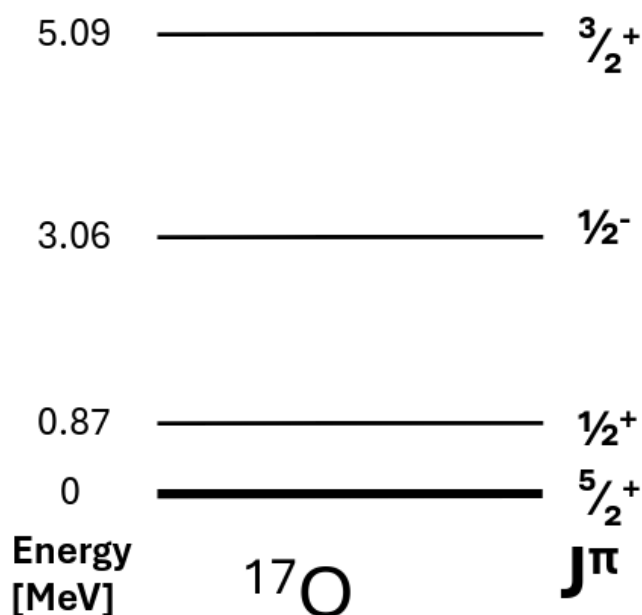
Draw the energy level sequence with spin-parities and energies for this 3 state system (there are two possibilities, full marks will be given for either one).

**[10]**

## SECTION 2

Answer **two** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

5. (a) The mass of the  $\pi^0$  meson is  $135 \text{ MeV}/c^2$ . From this, estimate the parameter  $a$  in the Woods-Saxon potential. [2]
- (b) Explain the contribution of heavier mesons in the nuclear exchange force. How does this inform the functional dependence of  $r$  on  $A$  in the Woods-Saxon potential? [3]
- (c) What is meant by magic numbers and where do they originate from? [2]
- (d) The energy spectrum of  $^{17}_8\text{O}$  is shown below:



The first energy levels of the nuclear shell model are:

$1s_{1/2}$ ,  $1p_{3/2}$ ,  $1p_{1/2}$ ,  $1d_{5/2}$ ,  $2s_{1/2}$  and  $1d_{3/2}$ .

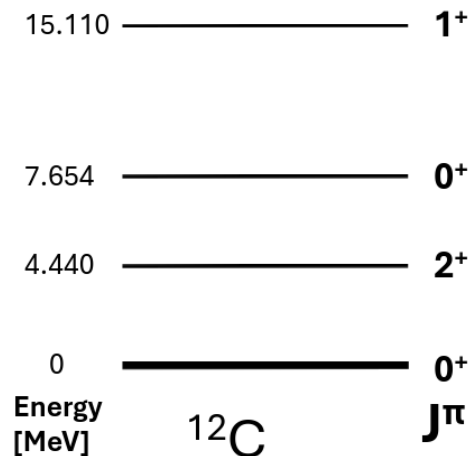
Assign the states in the diagram above to your simple shell model configurations and justify the energy ordering by:

- i. Drawing the shell model configuration for each possibility, what states can be produced by excitation of the valence neutron and why are they energy ordered in this way? [8]

**ANY CALCULATOR**

- ii. Drawing the shell model configuration, how can breaking a pair of protons give rise to the  $J^\pi = \frac{1}{2}^-$  state and what does this mean for the alignment of the np-pair in the same orbital? **[4]**
- iii. Explain in the mirror nucleus  $^{17}_9\text{F}$ , whether these same states will be higher or lower energy, and why. **[2]**
- iv. Draw the shell model configuration for the isobaric analogue state of the  $^{17}_7\text{N}$  ( $J^\pi = \frac{1}{2}^+$ ) ground state in  $^{17}_8\text{O}$ . **[4]**
- (e) For the odd-odd nucleus  $^{18}_9\text{F}$ , what are the spin-parities of the low-lying states expected from the  $1d_{5/2} \otimes 1d_{5/2}$  coupling? **[5]**

6. The energy spectrum of  $^{12}\text{C}$  is shown below.



The masses of  $^4_2\text{He}$ ,  $^8_4\text{Be}$  and  $^{12}_6\text{C}$  are 4.002603 u, 8.0053051 u and 12 u respectively. Consider only  $\alpha$  and  $\gamma$  decay modes for this question.

- (a) Calculate the Q-value for  $\alpha$  decay of  $^{12}\text{C}$  to  $^8\text{Be}$ . [2]
- (b) Explain how the first-excited state at 4.44 MeV will decay and estimate its lifetime. [4]
- (c) Explain how the second-excited state at 7.654 MeV will decay and estimate its lifetime by:
- Listing all the possible decay paths for this state and those that are forbidden (explaining why). [3]
  - Calculating the half-life for  $\alpha$  decay to the  $^8_4\text{Be}$  ground state. [10]
  - Calculating the half-life for all available  $\gamma$  decays. [2]
- (d) Explain how the high-lying  $J^\pi = 1^+$  state will decay, ignoring any possible excited states of  $^8\text{Be}$  and estimate the lifetime. [6]
- (e) Explain how one can determine the spin-parity of the second-excited state at 7.654 MeV by observing its decay mode. [3]

7. (a) i. The density of states term in the Fermi Golden Rule counts the number of final states. Show that for  $\beta$  decay:

$$\frac{dN}{dE_f} = p_e^2 \frac{V^2 (Q - T_e)^2}{4\pi^4 c^3 \hbar^6} dp_e,$$

[9]

- ii. What assumptions are made when calculating the transition matrix element for  $\beta$ -decay using Fermi's Golden Rule? [4]

- iii. The Q-value for the  $\beta^-$  of  $^{64}\text{Cu}$  is +0.580 MeV and the Q-value for **electron capture** is +1.675 MeV. On the same graph, draw the expected  $\beta^+$  and  $\beta^-$  intensity as a function of energy, with the endpoint energy for each calculated. [4]

- (b) i. The low-energy cross section for neutron-proton scattering is:

$$\sigma \approx \frac{4\pi}{k^2} (1 + ak)^2,$$

based on the wavenumber  $k$  **outside** the nuclear potential and the width of the nuclear potential  $a$ .

Given the n-p singlet cross section is 67.8 b, estimate the energy of the unbound singlet state in the deuteron, stating any assumptions made. [7]

- ii. In the low-energy limit, what will be the angular dependence of the differential cross section? [2]

- iii. Derive a formula for the phase shift,  $\delta$ , that occurs for n-p scattering by applying the usual boundary conditions for the finite square well potential. The form should depend only on  $k'$  (the wavenumber inside the well),  $k$  (the wavenumber outside the well) and the width of the nuclear potential,  $a$ . [4]

# Useful Formulae for Nuclear Physics

Coulomb Barrier

$$V_{\text{Coul}}(r) = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 R}, \quad \text{where } R = 1.2A^{1/3} \text{ fm} \quad (1)$$

Radial Schrödinger Equation

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[ V(r) + \frac{l(l+1)\hbar^2}{2mr^2} \right] u(r) = E u(r), \quad \text{where } R(r) = \frac{u(r)}{r}$$

Deuteron Wavefunction Matching Condition

$$k_1 \cot(k_1 R) = -k_2, \quad \text{where } k_1 = \sqrt{\frac{2m(E + V_0)}{\hbar^2}}, \quad k_2 = \sqrt{-\frac{2mE}{\hbar^2}}$$

Magnetic Moment

$$\langle \mu \rangle = \frac{\{g_l[j(j+1) + l(l+1) - s(s+1)] + g_s[j(j+1) + s(s+1) - l(l+1)]\} \mu_N}{2(j+1)}$$

Alpha-decay Half-life

$$t_{1/2} = \ln(2) \frac{R}{c} \sqrt{\frac{mc^2}{2(V_0 + Q)}} \exp\{2G\}, \quad \text{where } G = \sqrt{\frac{2m}{\hbar^2 Q}} \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0} \left[ \frac{\pi}{2} - 2 \left( \frac{Q}{B} \right)^{1/2} \right]$$

Beta-decay  $ft$  Values

$$f(Z, T_e) t_{1/2} = \ln(2) \frac{2\pi^3 \hbar^7}{g_F^2 |M_{fi}|^2 m_e^5 c^4}$$

Internal Conversion Coefficients

$$\alpha(ML) \approx \frac{Z^3}{n^3} \left( \frac{e^2}{4\pi\epsilon_0 \hbar c} \right)^4 \left( \frac{2m_e c^2}{E} \right)^{L+3/2}$$

$$\alpha(EL) \approx \frac{Z^3}{n^3} \left( \frac{L}{L+1} \right) \left( \frac{e^2}{4\pi\epsilon_0 \hbar c} \right)^4 \left( \frac{2m_e c^2}{E} \right)^{L+5/2}$$

Weisskopf Estimates (for  $E_\gamma$  in MeV)

|                                                       |                                                            |
|-------------------------------------------------------|------------------------------------------------------------|
| $\lambda(E1) = 1.0 \times 10^{14} A^{2/3} E_\gamma^3$ | $\lambda(M3) = 1.1 \times 10^1 A^{4/3} E_\gamma^7$         |
| $\lambda(M1) = 3.1 \times 10^{13} E_\gamma^3$         | $\lambda(E4) = 1.1 \times 10^{-5} A^{8/3} E_\gamma^9$      |
| $\lambda(E2) = 7.4 \times 10^7 A^{4/3} E_\gamma^5$    | $\lambda(M4) = 3.3 \times 10^{-6} A^2 E_\gamma^9$          |
| $\lambda(M2) = 2.2 \times 10^7 A^{2/3} E_\gamma^5$    | $\lambda(E5) = 2.4 \times 10^{-12} A^{10/3} E_\gamma^{11}$ |
| $\lambda(E3) = 3.5 \times 10^1 A^2 E_\gamma^7$        | $\lambda(M5) = 7.4 \times 10^{-13} A^{8/3} E_\gamma^{11}$  |



## Physical Constants and Units

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|                                                        |                                      |                                                                                          |
|--------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------|
| Acceleration due to gravity                            | $g$                                  | $9.81 \text{ m s}^{-2}$                                                                  |
| Gravitational constant                                 | $G$                                  | $6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$                                    |
| Ice point                                              | $T_{ice}$                            | $273.15 \text{ K}$                                                                       |
| Avogadro constant                                      | $N_A$                                | $6.022 \times 10^{23} \text{ mol}^{-1}$                                                  |
| [ <i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i> ] |                                      |                                                                                          |
| Gas constant                                           | $R$                                  | $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$                                                |
| Boltzmann constant                                     | $k, k_B$                             | $1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$    |
| Stefan constant                                        | $\sigma$                             | $5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$                                   |
| Rydberg constant                                       | $R_\infty$                           | $1.097 \times 10^7 \text{ m}^{-1}$                                                       |
|                                                        | $R_\infty hc$                        | $13.606 \text{ eV}$                                                                      |
| Planck constant                                        | $h$                                  | $6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$            |
|                                                        | $h/2\pi$                             | $\hbar$<br>$1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$ |
| Speed of light <i>in vacuo</i>                         | $c$                                  | $2.998 \times 10^8 \text{ m s}^{-1}$                                                     |
|                                                        | $\hbar c$                            | $197.3 \text{ MeV fm}$                                                                   |
| Charge of proton                                       | $e$                                  | $1.602 \times 10^{-19} \text{ C}$                                                        |
| Mass of electron                                       | $m_e$                                | $9.109 \times 10^{-31} \text{ kg}$                                                       |
| Rest energy of electron                                |                                      | $0.511 \text{ MeV}$                                                                      |
| Mass of proton                                         | $m_p$                                | $1.673 \times 10^{-27} \text{ kg}$                                                       |
| Rest energy of proton                                  |                                      | $938.3 \text{ MeV}$                                                                      |
| One atomic mass unit                                   | $u$                                  | $1.66 \times 10^{-27} \text{ kg}$                                                        |
| Atomic mass unit energy equivalent                     |                                      | $931.5 \text{ MeV}$                                                                      |
| Electric constant                                      | $\epsilon_0$                         | $8.854 \times 10^{-12} \text{ F m}^{-1}$                                                 |
| Magnetic constant                                      | $\mu_0$                              | $4\pi \times 10^{-7} \text{ H m}^{-1}$                                                   |
| Bohr magneton                                          | $\mu_B$                              | $9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$                                  |
| Nuclear magneton                                       | $\mu_N$                              | $5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$                                  |
| Fine-structure constant                                | $\alpha = e^2/4\pi\epsilon_0\hbar c$ | $7.297 \times 10^{-3} = 1/137.0$                                                         |
| Compton wavelength of electron                         | $\lambda_c = h/m_e c$                | $2.426 \times 10^{-12} \text{ m}$                                                        |
| Bohr radius                                            | $a_0$                                | $5.2918 \times 10^{-11} \text{ m}$                                                       |
| angstrom                                               | $\text{\AA}$                         | $10^{-10} \text{ m}$                                                                     |
| barn                                                   | $\text{b}$                           | $10^{-28} \text{ m}^2$                                                                   |
| torr (mm Hg at 0 °C)                                   | torr                                 | $133.32 \text{ Pa (N m}^{-2}\text{)}$                                                    |

**Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.**

**Important Reminders**

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

**Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.**